

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – [Smart Study Generator Agent]

Domain of Project – [Education]

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Problem Statement -

Problem Statement : Smart Study Generator Agent

The Challenge

Students today face an overwhelming flood of information—lecture notes, textbooks, research papers, YouTube tutorials, and online resources. This content is scattered, unorganized, and often not personalized to their learning pace or style. As a result, students struggle with finding the right study material, summarizing key concepts, and staying exam-ready. Traditional study tools only provide static notes or flashcards, failing to adapt dynamically to student needs. The challenge is to move beyond static resources toward an intelligent, adaptive study companion.

The Objective

Build an agentic AI-powered Smart Study Generator that personalizes and streamlines learning for students. The solution should:

- 1. Fuse Learning Resources:** Aggregate content from class notes, textbooks, research articles, and online tutorials into structured, easy-to-digest study material.
- 2. Enable Multimodal Input:** Allow students to upload notes, voice queries, or images (e.g., textbook pages), and generate concise summaries, flashcards, or concept maps.
- 3. Create a Predictive & Agentic Layer:** Recommend personalized study plans, highlight important topics based on syllabus or past exams, and provide predictive alerts (e.g., “Focus on Topic X—frequently asked in exams”).
- 4. Visualize Study Progress:** Offer a real-time dashboard showing learning milestones, weak areas, practice quizzes, and performance forecasts

Proposed Solution -

Proposed Solution: StudyGen AI – AI-Powered Smart Study Assistant

StudyGen AI is an AI-powered learning platform that helps students study efficiently using Agentic AI and Retrieval-Augmented Generation (RAG). It collects information from textbooks, lecture notes, PDFs, and trusted educational resources to provide personalized learning support.

Key Features:

- AI-powered chapter summaries
- PDF and image upload with document analysis
- Intelligent doubt-solving chatbot
- Automatic quiz and flashcard generation
- Personalized study plans and revision schedules
- Student progress tracking dashboard
- Important topic and exam question prediction

How It Works

Student uploads notes, PDFs, or images. AI extracts and understands the content. RAG retrieves relevant information from the knowledge base. AI generates summaries, quizzes, flashcards, and study plans. Students interact with the chatbot for personalized learning.

Benefits:

- Saves study time
- Improves exam preparation
- Personalized learning experience

Technology Used

- **IBM Granite model** - for eg – ibm-granite-3-2-8b – For natural language understanding, reasoning, and personalization
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Nutrition Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Nutrition Agent.
- **RAG**– RAG to **fetch real data first, then generate intelligent answers**, making the Nutrition Agent more trustworthy and effective.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance
- **Vector Database** (FAISS/Chroma) – For RAG-based nutritional data retrieval

Model workflow -

IBM watsonx Orchestrate

EnglishContact SalesBuy Now29 trial days left

Home / Manage agents / StudyGen AI

Test agentDeploy

StudyGen AI

Model: GPT-OSS 120B – Open...

Profile

Knowledge

Toolset

Behavior

Channels Preview

Knowledge sources

Replace source +

Study Material Knowledge Base

This knowledge base contains textbooks, lecture notes, PDFs, research papers, study guides, syllabus documents, and educational resources. The agent uses this knowledge to answ...

Used by 1 agent

Edited 4 hours ago

Preview

Running in debug mode

Welcome to Study gen AI

Accuracy of generated answers may vary. Please double-check responses.

Predict the most important topics for the exam

Find my weak topics and suggest improvements

1/3

Type something...

IBM watsonx Orchestrate

EnglishContact SalesBuy Now29 trial days left

Home / Manage agents / StudyGen AI

Test agentDeploy

StudyGen AI

Model: GPT-OSS 120B – Open...

Profile

Knowledge

Toolset

Behavior

Channels Preview

Behavior

Define how and where your agent should react to requests and respond to users.

Instructions

Restrictions:

- Do not provide false or misleading information.

- Do not answer harmful or illegal requests.

- For medical, legal, or financial questions, advise users to consult qualified professionals.

- Stay focused on educational and learning-related assistance.

Specify what your agent should do, how it should respond and any restrictions it should follow.

Guidelines

Add rules that guide an agent's actions and responses.

Simple Explanation

Chapter Summary

Exam Preparation

Quiz Generator

Preview

Running in debug mode

2:31 PM

Welcome to Study gen AI

Accuracy of generated answers may vary. Please double-check responses.

Predict the most important topics for...

Find my weak topics and suggest...

Create a quiz based on the uploaded PDF.

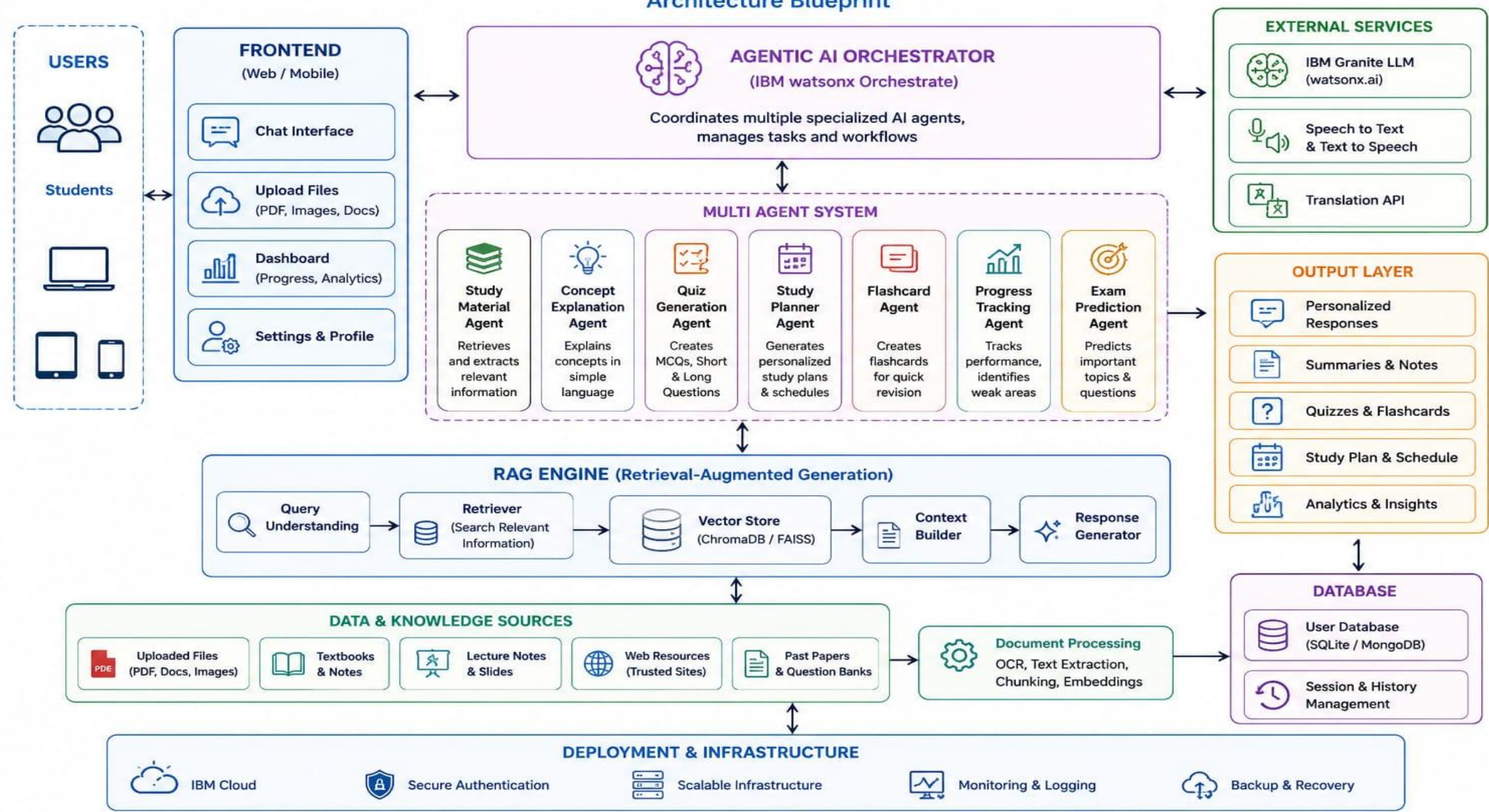
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Type something...

Architecture blueprint

STUDYGEN AI – Smart Study Generator Agent

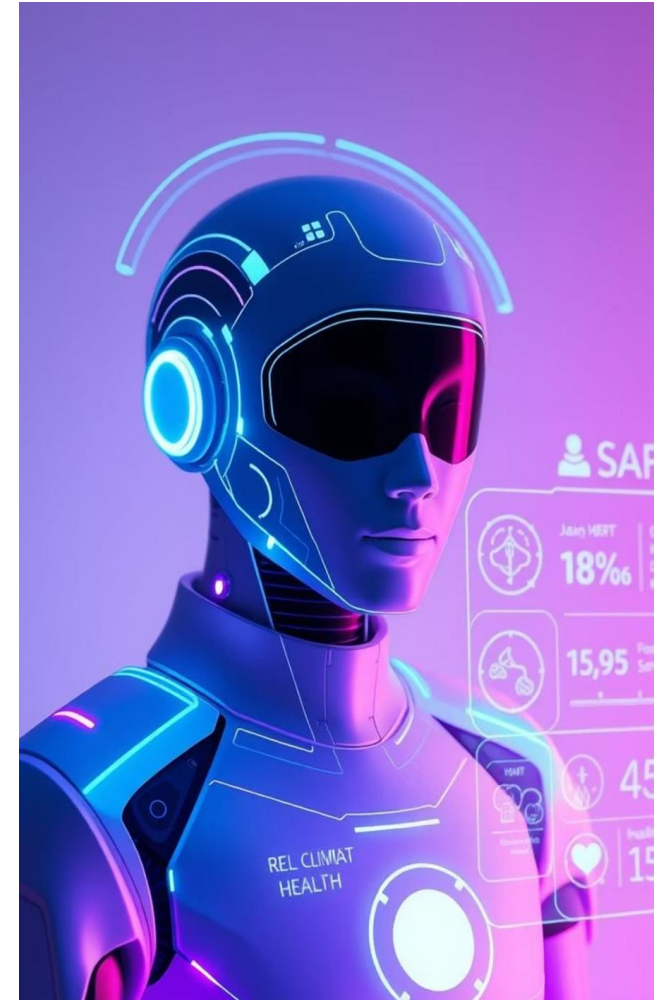
Architecture Blueprint



Role of Agentic AI in the solution

Role of Agentic AI in Smart Study Generator

Agentic AI is the **core intelligence** behind StudyGen AI, enabling the system to perform multiple learning tasks through a team of specialized AI agents. Instead of functioning as a simple chatbot, Agentic AI coordinates different agents to retrieve information, analyze study materials, and provide personalized learning support. The Study Material Agent uses **Retrieval-Augmented Generation (RAG)** to search uploaded PDFs, notes, and textbooks for relevant information. The Concept Explanation **Agent simplifies complex topics** into easy-to-understand explanations with examples, while the Quiz Generation Agent creates practice questions, MCQs, and flashcards to reinforce learning. The Study Planner Agent **develops personalized study schedules** based on the student's exam dates and available study time, and the Progress Tracking Agent monitors learning progress, identifies weak areas, and recommends improvements. By enabling these agents to collaborate intelligently, **Agentic AI delivers personalized, interactive, and context-aware learning**, helping students study more efficiently, improve their understanding, and achieve better academic performance.



Novelty and Uniqueness

Personalized AI Learning: Creates customized study plans based on the student's syllabus, exam date, and available study time.

Multi-Source Learning: Combines information from textbooks, notes, PDFs, research papers, and trusted online resources using RAG.

Interactive AI Tutor: Goes beyond answering questions by explaining concepts, generating quizzes, flashcards, and revision notes.

Exam-Focused Assistance: Predicts important topics and generates likely exam questions to support targeted preparation.

Multimodal Support: Accepts text, PDF, and image inputs, making it easy to study from different types of learning materials.

Progress Tracking: Monitors learning progress, identifies weak areas, and recommends personalized improvements through a dashboard

Git Hub Link

Github URL –<https://github.com/Shra-ddha21-git/studygen-ai-agent.git>

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Future Scope

Study Gen AI has significant potential for future enhancement by incorporating advanced Artificial Intelligence, Agentic AI, and multimodal learning capabilities. In the future, the platform can support voice-based interactions and multilingual learning, allowing students to study in their preferred language. It can integrate with Learning Management Systems (LMS) such as Moodle and Google Classroom to automatically access course materials and assignments. The system can also provide real-time collaboration, enabling students to participate in group discussions and AI-assisted study sessions.

Additionally, Study Gen AI can include AI-powered career guidance, personalized course recommendations, and interview preparation based on students' academic performance. Integration with Augmented Reality (AR) and Virtual Reality (VR) can make learning more interactive through immersive educational experiences. Advanced learning analytics can offer detailed insights into student progress, predict performance, and recommend targeted improvement strategies. By continuously evolving with emerging AI technologies, StudyGen AI can become a comprehensive personalized learning ecosystem that enhances education, improves academic outcomes, and supports lifelong learning.

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In recognition of the commitment to achieve professional excellence



Shraddha Tiwari

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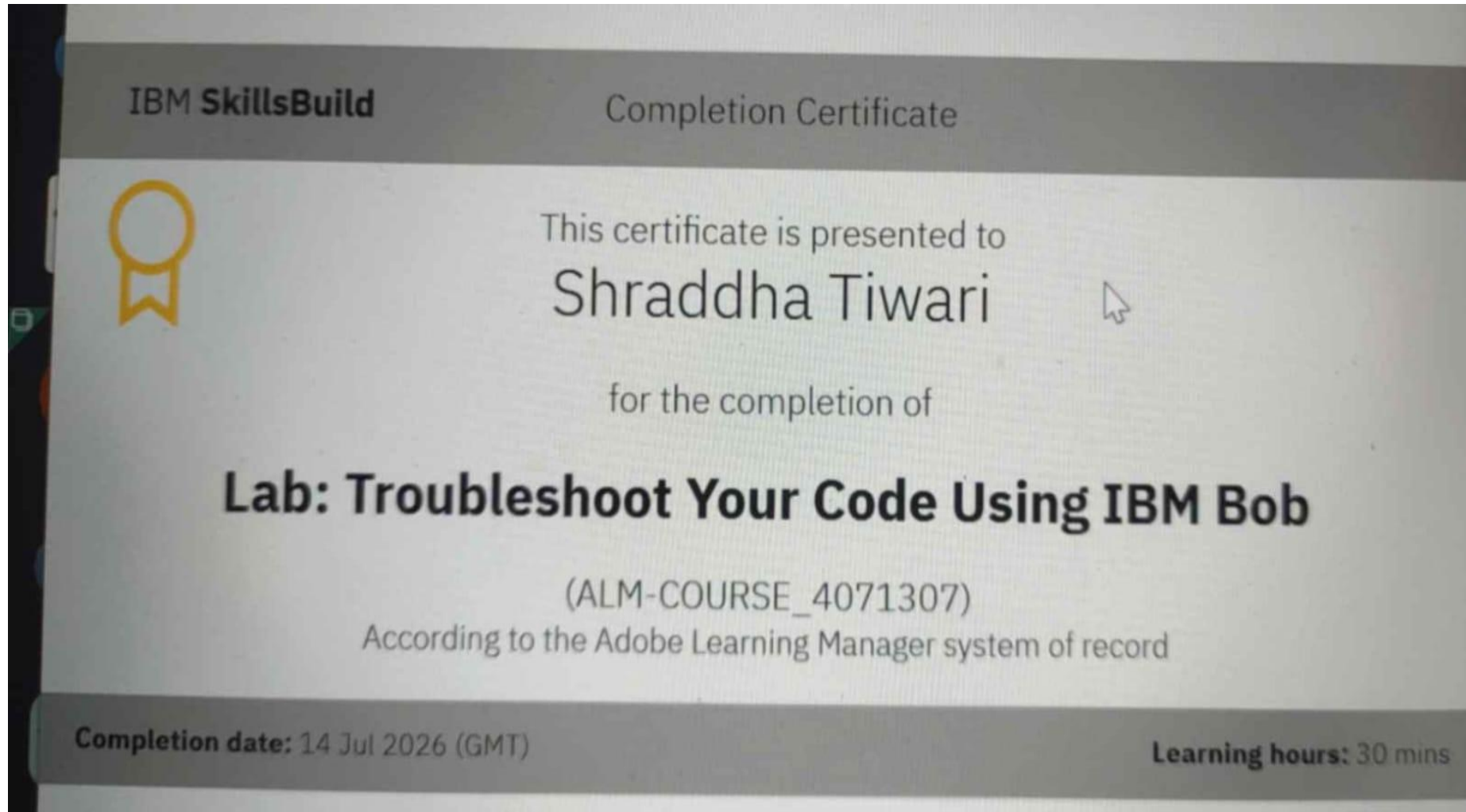
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